

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States both hypotheses using correct language	AO2.5	B1	$H_0: \lambda = 50$ $H_1: \lambda < 50$
	Selects and uses Poisson model with $\lambda = 50$ to find $P(\text{total vehicles} \leq 30)$ or $P(\text{total vehicles} < 30)$ or $P(\text{total vehicles} \leq 33)$ and $P(\text{total vehicles} \leq 34)$	AO3.3	M1	$X \sim \text{Po}(50)$ $P(X \leq 30) = 0.002$ $(P(X \leq 33) = 0.007 \text{ and } P(X \leq 34) = 0.0108)$ $p\text{-value} = 0.002 < 0.01$ $(30 < 34 = \text{critical value})$
	Obtains AWRP $P(X \leq 30) = 0.002$ (0.001594) or AWRP $P(X \leq 33) = 0.007$ (0.0069788) and AWRP $P(X \leq 34) = 0.0108$ (0.01078145)	AO1.1b	A1	Reject H_0 in favour of H_1 Significant evidence to suggest that the mean number of vehicles passing the service station per minute has reduced
	Evaluates the Poisson model by comparing 'their' p -value with 0.01 or 30 with their critical value	AO3.5a	R1	
	Infers H_0 rejected	AO2.2b	E1	
	Concludes in context (not definite)	AO3.2a	E1	
(b)	States meaning in context of Type I error	AO3.2a	E1	Type I error is to conclude that the mean number of vehicles passing the service station per minute using the junction has reduced when it has not.
(c)	States that rate at which events occur is unlikely to be constant over time in context or States that the events do not occur independently in context Must be consistent	AO3.5b	E1	The rate at which vehicles pass the service station is unlikely to be constant over time. or Vehicles may not pass the service station independently, eg convoy
				Total 8 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States both hypotheses using correct language	AO2.5	B1	$H_0: \lambda = 14$ $H_1: \lambda < 14$ 1 tail 5 %

			Allow $\lambda = 3.5$ used
Selects and uses Poisson model with $\lambda = 14$ to find $P(\text{total sales} \leq 8)$ or $P(\text{total sales} < 8)$	AO3.3	M1	$T \sim \text{Po}(14)$
Obtains 0.062	AO1.1b	A1	$P(\text{Total sales} \leq 8) = 0.062$
Evaluates the Poisson model by comparing 'their' p -value with 0.10	AO3.5a	M1F	$p\text{-value} = 0.062 \quad 0.062 > 0.05$
Infers H_0 accepted	AO2.2b	E1	Accept H_0
Concludes in context (not definite)	AO3.2a	E1	No significant evidence to suggest that Keith's mean number of properties sold per week in spring has reduced.
(b) States meaning in context of Type II error	AO3.2a	E1	Type II error is to conclude that there is no significant evidence to suggest that Keith's mean number of properties sold per week in spring has reduced when, in fact, the mean number sold has reduced.
(c) Advantage clearly stated	AO1.2	E1	Advantage Less likely to reject H_0 when H_0 is, in fact, true or Less likely to make a Type I error
Disadvantage clearly stated	AO1.2	E1	Disadvantage Less likely to accept H_1 when H_0 is, in fact, false or More likely to make a Type II error
Total 9 marks			

Q3.

$X_D \sim \text{Po}(24)$

- (a) $H_0 : \lambda \text{ (or mean)} = 2 \text{ (or 10)}$
 $H_1 : \lambda \text{ (or mean)} > 2 \text{ (or 10)}$

Both; or equivalent

B1

$$P(Y \geq 17) = 1 - P(Y \leq 16)$$

Accept $1 - P(Y \leq 17)$

M1

$$= 1 - 0.0.9730 = 0.027$$

AWRT

A1

< 0.10 (10%)

$$[z = 2.05 \text{ to } 2.38 > 1.2816]$$

Comparison of probability with 0.1

Comparison of z with 1.2816 or 1.6449

M1

Thus evidence, at 10% level, of increase
in mean daily number of requests

F on probability or on z

A1F

5

- (b) CV of Y is such that $P(Y \geq CV) \leq 0.10$ (10%)

Can be implied by 13, 14 or 15

Accept $P(Y = CV) = 0.10$

M1

Thus $P(Y \leq CV - 1) \geq 0.90$

Can be implied by 13, 14 or 15

Accept $P(Y = CV) = 0.90$

M1

Thus CV = 15

CAO

A1

3

- (c) Power = $1 - P(\text{Type II error})$

Or equivalent

$$= 1 - P(\text{accept } H_0 \mid H_0 \text{ false})$$

$$= P(\text{accept } H_1 \mid H_1 \text{ true})$$

Stated or implied use

B1

$$\lambda = 5 \times 3 = 15$$

Stated or implied use of Po(15)

B1

Thus power = $P(Y \geq CV)$

Attempt at a probability based on C's CV from (b) and Po(15)

M1

$$= P(Y \geq 15) = 1 - P(Y \leq 14)$$

$$= 1 - 0.4657 = 0.53 \text{ to } 0.54$$

AWFW (0.5343)

A1

4

[12]

Q4.

(a) $H_0 : \lambda = 13$

CAO; OE

B1

$$H_1 : \lambda < 13$$

CAO; OE

B1

$$P(R \leq 10 \mid \text{Po}(13))$$

Used or implied

M1

$$= 0.2517$$

AWFW 0.251 to 0.252

A1

Prob of $0.2517 > 0.10$ (10%)

$$z = -0.83 \text{ to } -0.70 > -1.28$$

Comparison of prob with 0.10

Comparison of z with -1.28

M1

Thus no evidence, at 10% level, of a reduction in the mean value of R

ft on probability or z

In 'context' and qualified

A1ft

6

- (b) Require $P(R \leq r \mid \text{Po}(13)) \approx 0.10$
Stated or implied

M1

Critical Region is $R \leq 8$ or $R < 9$

Accept $R = 8$

May be scored in (a)

A1

2

- (c) Require $P(\text{accept } H_0 \mid H_0 \text{ false})$
OE; PI

B1

$$= P(R > 8 \mid \text{Po}(6.5))$$

Use of Po(6.5)

M1

$$= 1 - P(R \leq 8 \mid \text{Po}(6.5))$$

m1

$$= 1 - 0.7916$$

$$= 0.208 \text{ to } 0.209$$

AWFW (0.2084)

A1

4

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

[12]

Q5.

$$H_0 : \mu = 7.0$$

$$H_1 : \mu < 7.0$$

B1

$X \sim$ number failing to turn up per day

$$\therefore X \sim \text{Po}(7.0)$$

AWRT 0.082

M1A1

$$P(X \leq 3) = 0.0818 \text{ (tables)}$$

$$\left. \begin{array}{l} > 0.05 \\ \therefore \text{accept } H_0 \end{array} \right\}$$

m1

insufficient evidence at the 5% level of significance to support the manager's claim

E1

[5]

Q6.

$\lambda = 0.4$ per pack

(a) $\lambda = 20 \times 0.4 = 8$

cao; can be awarded in (b) for 40

B1

$H_0: \lambda = 0.4$ (or 8)

$H_1: \lambda > 0.4$ (or 8)

both; can be awarded in (b) for 0.4 (or 40); accept μ , not p

B1

$P(X \geq 13) = 1 - P(X \leq 12)$

use of which Poisson; accept

$1 - P(X \leq 13)$ or $1 - 0.9658$ or 0.0342

M1

$= 1 - 0.9362 = 0.064$

awrt

A1

($> 5\%$)

Thus, at 5% level, insufficient evidence to support the claim that λ has increased

ft on p-value andd 5%

A1ft

5

(b) $\lambda = 100 \times 0.4 = 40$

$\therefore$ Normal approximation with

use of

M1

mean (μ) = 40 and variance (σ^2) = 40

cao both; accept $\sigma = 6.3$ awrt;

accept 0.4 & 0.004 (0.063) cao both

A1

SL $\alpha = 0.05$

CV $Z = + 1.6449$

*awfw +1.64 to + 1.65;
or 0.05 cao for p-value*

B1

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \text{ or } \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

standardisation of (51.5, 52, 52.5 or 0.52)

*with their μ and their σ or their $\frac{\sigma}{\sqrt{100}}$,
(not σ^2); accept $(\mu - x)$;*

M1

$$= \frac{(51.5 \text{ or } 52) - 40}{\sqrt{40}} \text{ or } \frac{0.52 - 0.4}{\sqrt{0.4}}$$

$$= 1.81 \text{ to } 1.91$$

*awfw
(p-value = 0.028 to 0.035 awfw)*

A1

Thus, at 5% level, sufficient evidence to support the claim that λ has increased

ft on z-value and CV, both positive or on p-value and 5%

A1ft

6

[11]

Q7.

(a) $\lambda = 1.8$ per day

$$\lambda = 5 \times 1.8 = 9$$

cao

B1

$$P(X \leq 6) = 0.2068$$

*attempt at ≤ 6 using Poisson
awfw 0.20 to 0.21*

M1 A1

0.10

comparison with 10% must be seen

m1

4

(Thus insufficient evidence)

ag

(b) $\lambda = 180 \times 1.8 = 324$

cao

B1

Normal approximation

use of

$$\mu = \sigma^2 = \lambda = 324$$

ft on λ (not 1.8 or 9)

M1

$$H_0 : \lambda = 1.8 \text{ (324)}$$

$$H_1 : \lambda < 1.8 \text{ (324)}$$

A1ft

both can be scored in (a)

B1

$$\alpha = 0.10$$

©2025 Exam Papers Practice. All rights reserved.

cv $Z = -1.2816$

awrt ± 1.28

B1

$$z = \frac{x - \mu}{\sigma}$$

use of

accept 296.5 or 297.5

M1

$$= \frac{297 - 324}{18} = -1.5$$

awfw ± 1.47 to ± 1.53

A1

Thus evidence, at 10% level, that mean daily number
is now less than 1.8

or equivalent

ft on z and cv,

providing consistent with regard to signs

A1 ✓

8

[12]



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Examiner reports

Q1.

In part (a), the majority of students scored at least three marks. Most students stated the correct hypotheses with some students using μ instead of λ . A minority of students stated hypotheses for a goodness of fit test or used the sample value in their hypotheses. Most students calculated the Poisson probability correctly. The most common error was to calculate the probability of exactly 30 rather than 30 and less. A significant proportion of students found the critical region instead but some students incorrectly calculated the region or did not provide enough evidence that they had found the critical region. When comparing the Poisson probability with the significance level, some students made errors in order of magnitude, either mistaking 0.00159 as 1.59% or 1% as 0.1 or 0.001. Some students lost the final mark for being too definite in their final conclusion or not writing a conclusion in context.

In part (b), most students correctly identified Type I error but many failed to do so in context. Some students described Type II error instead or described the probability of Type I error. A small amount of students were not careful with the use of double negatives in their language.

The majority of students did not score the mark for part (c). Many students criticised the time that Xander took the sample but did not relate this to the Poisson model assumption of a constant rate over time. A significant proportion of students declared that the process was not random whilst giving descriptions in context of the events not being independent.

Q3.

As stated earlier, much of this question proved too much of a challenge for most candidates. In part (a), many candidates failed to state hypotheses in terms of λ . Also, despite the wording of the question, about 50% of candidates attempted an approximate normal test based on either $\lambda = 2$ or $\lambda = 10$. The first is simply incorrect and the second is at best borderline. In general candidates should not opt for approximate tests for binomial proportions or Poisson means when tables in the supplied booklets are available for exact tests.

©2025 Exam Papers Practice. All rights reserved.

Only candidates opting for an exact test in part (a) stood any realistic chance of scoring marks in part (b). Most such candidates had a basic idea as to what was needed but only the very best obtained the correct value of 15. The usual alternative was 14, but 13 and 16 were seen. In part (c), most candidates defined power correctly, usually as $1 - P(\text{accept } H_0 \mid H_0 \text{ false})$. They then attempted to calculate this probability using $\lambda = 15$ and so often scored at least 2 marks. A very small proportion of candidates are to be congratulated on scoring all 17 marks in correctly answering this final question.

Q4.

In part (a), candidates were expected to use an 'exact' test since Table 2 tabulates $P(X \leq x \mid \lambda = 13)$. Those who did often scored full marks whereas those who attempted a normal approximation sometimes introduced further inaccuracies. Only the best candidates, usually those who used an exact test in part (a), scored marks in part (b). The same candidates often then scored well in part (c). Those candidates determined to use a normal approximation at all costs usually opted for $N(6.5, 6.5)$ or even $N(6.5, 13)$, neither of which was accepted.

Q5.

Candidates found this to be the most difficult question on the paper. In spite of this, many excellent solutions were obtained by a variety of methods. However, conclusions were either too positive in nature or not given in context.

The hypotheses were usually stated correctly. Unfortunately, in many cases, candidates either failed to consider $X \leq 3$ or used the normal distribution in their attempt to calculate $P(X \leq 3)$.

Q6.

It was pleasing to see the overall improvement over previous papers in attempts at this type of question.

In part (a), hypotheses were usually stated correctly and were followed by attempts at $P(X \geq 13)$ for $Po(8)$ from tables; here marks were not awarded for using a normal approximation. All too often this probability was equated to $1 - P(X \geq 13) = 0.0342$ instead of $1 - P(X \geq 12) = 0.0638$. For both sets of candidates, about 50% then came to an incorrect conclusion through statements of the form “ $0.0638 > 0.05 \Rightarrow \text{Reject } H_0$ ” or “ $0.0342 < 0.05 \Rightarrow \text{Accept } H_0$ ”; perhaps confused with comparisons using critical values.

Q7.

In part (a), many candidates attempted $P(X \geq 6)$ for $Po(1.8)$ rather than $P(X \leq 6)$ for $Po(9)$. As a result, conclusions to match the quoted answer were ‘fudged’. Such candidates generally persisted with $\lambda = 1.8$ in part (b) by attempting a normal approximation based on

$$z = \frac{1.65\left(\frac{297}{180}\right) - 1.8}{\sqrt{1.8}}$$

a value of $\frac{1.65\left(\frac{297}{180}\right) - 1.8}{\sqrt{1.8}}$. Such an approach requires a denominator of $\sqrt{\frac{1.8}{180}}$. The many candidates who used the normal approximation for $Po(324)$ generally scored most, if not all, of the subsequent eight marks. Rather than compare -1.5 with -1.2816 , a considerable number of candidates continued with the equivalent method used in part (a) and so compared correctly 0.067 with 0.10 .

©2025 Exam Papers Practice. All rights reserved.